

TABLA DE INTEGRALES

1. $\int k f(u) du = k \int f(u) du$	49. $\int u^2 \cos u du = 2u \cos u + (u^2 - 2) \sin u + c$
2. $\int [f(u) + g(u) + \dots] du = \int f(u) du + \int g(u) du + \dots$	50. $\int u^n \sin u du = -u^n \cos u + n \int u^{n-1} \cos u$
3. $\int u dv = uv - \int v du$	51. $\int u^n \cos u du = u^n \sin u - n \int u^{n-1} \sin u$
4. $\int du = u + c$	52. $\int \sin mu \sin nu du = -\frac{\sin(m+n)u}{2(m+n)} + \frac{\sin(m-n)u}{2(m-n)} + c$
5. $\int k du = ku + c$	53. $\int \sin mu \cos nu du = -\frac{\cos(m+n)u}{2(m+n)} - \frac{\cos(m-n)u}{2(m-n)} + c$
6. $\int u^n du = \frac{1}{n+1} u^{n+1} + c; n \neq -1$	54. $\int \cos mu \cos nu du = \frac{\sin(m+n)u}{2(m+n)} + \frac{\sin(m-n)u}{2(m-n)} + c$
7. $\int u^{-1} du = \int \frac{1}{u} du = \ln u + c$	55. $\int \sin^m u \cos^n u du = -\frac{\sin^{m-1} u \cos^{n+1} u}{m+n} + \frac{m-1}{m+n} \int \sin^{m-2} u \cos^n u du$
8. $\int e^u du = e^u + c$	56. $\int \sin^{-1} u du = u \sin^{-1} u + \sqrt{1-u^2} + c$
9. $\int a^u du = \frac{1}{\ln a} a^u + c$	57. $\int \cos^{-1} u du = u \cos^{-1} u - \sqrt{1-u^2} + c$
10. $\int u e^u du = e^u (u - 1) + c$	58. $\int \tan^{-1} u du = u \tan^{-1} u - \ln \sqrt{1+u^2} + c$
11. $\int u^n e^u du = u^n e^u - n \int u^{n-1} e^u$	59. $\int \cot^{-1} u du = u \cot^{-1} u + \ln \sqrt{1+u^2} + c$
12. $\int u^n a^u du = \frac{u^n a^u}{\ln a} - \frac{n}{\ln a} \int u^{n-1} a^u$	60. $\int \sec^{-1} u du = u \sec^{-1} u - \ln u + \sqrt{u^2 - 1} + c$ $= u \sec^{-1} u - \cosh^{-1} u + c$
13. $\int \frac{e^u}{u^n} du = -\frac{e^u}{(n-1)u^{n-1}} + \frac{1}{n-1} \int \frac{e^u}{u^{n-1}} du$	61. $\int \csc^{-1} u du = u \csc^{-1} u + \ln u + \sqrt{u^2 - 1} + c$ $= u \csc^{-1} u + \cosh^{-1} u + c$
14. $\int \frac{a^u}{u^n} du = -\frac{a^u}{(n-1)u^{n-1}} + \frac{\ln a}{n-1} \int \frac{a^u}{u^{n-1}} du$	62. $\int u \sin^{-1} u du = \frac{2u^2 - 1}{4} \sin^{-1} u + \frac{u \sqrt{1-u^2}}{4} + c$
15. $\int \ln u du = u \ln u - u + c$	63. $\int u \cos^{-1} u du = \frac{2u^2 - 1}{4} \cos^{-1} u - \frac{u \sqrt{1-u^2}}{4} + c$
16. $\int u^n \ln u du = \frac{u^{n+1}}{(n+1)^2} ((n+1) \ln u - 1) + c$	64. $\int u \tan^{-1} u du = \frac{u^2 + 1}{2} \tan^{-1} u - \frac{u}{2} + c$
17. $\int \frac{1}{u \ln u} du = \ln \ln u + c$	65. $\int u^n \sin^{-1} u du = \frac{1}{n+1} \left(u^{n+1} \sin^{-1} u - \int \frac{u^{n+1} du}{\sqrt{1-u^2}} \right),$ $n \neq -1$
18. $\int \sin u du = -\cos u + c$	66. $\int u^n \cos^{-1} u du = \frac{1}{n+1} \left(u^{n+1} \cos^{-1} u + \int \frac{u^{n+1} du}{\sqrt{1-u^2}} \right),$ $n \neq -1$
19. $\int \cos u du = \sin u + c$	
20. $\int \tan u du = \ln \sec u + c$	
21. $\int \cot u du = \ln \sin u + c$	
22. $\int \sec u du = \ln \sec u + \tan u + c$	
23. $\int \csc u du = \ln \csc u - \cot u + c$	

24. $\int \sin^2 u du = \frac{1}{2}u - \frac{1}{4}\sin 2u + c$
25. $\int \cos^2 u du = \frac{1}{2}u + \frac{1}{4}\sin 2u + c$
26. $\int \tan^2 u du = \tan u - u + c$
27. $\int \cot^2 u du = -\cot u - u + c$
28. $\int \sec^2 u du = \tan u + c$
29. $\int \csc^2 u du = -\cot u + c$
30. $\int \sin^3 u du = -\frac{1}{3}(2 + \sin^2 u) \cos u + c$
31. $\int \cos^3 u du = \frac{1}{3}(2 + \cos^2 u) \sin u + c$
32. $\int \tan^3 u du = \frac{1}{2}\tan^2 u + \ln |\cos u| + c$
33. $\int \cot^3 u du = -\frac{1}{2}\cot^2 u - \ln |\sin u| + c$
34. $\int \sec^3 u du = \frac{1}{2}\sec u \tan u + \frac{1}{2}\ln |\sec u + \tan u| + c$
35. $\int \csc^3 u du = -\frac{1}{2}\csc u \cot u + \frac{1}{2}\ln |\csc u - \cot u| + c$
36. $\int \sec u \tan u du = \sec u + c$
37. $\int \csc u \cot u du = -\csc u + c$
38. $\int e^{au} \sin nu du = \frac{e^{au}}{a^2 + n^2}(a \sin nu - n \cos nu) + c$
39. $\int e^{au} \cos nu du = \frac{e^{au}}{a^2 + n^2}(a \cos nu + n \sin nu) + c$
40. $\int \sin^n u du = -\frac{1}{n}\sin^{n-1} u \cos u + \frac{n-1}{n} \int \sin^{n-2} u du$
41. $\int \cos^n u du = \frac{1}{n}\cos^{n-1} u \sin u + \frac{n-1}{n} \int \cos^{n-2} u du$
42. $\int \tan^n u du = \frac{1}{n-1}\tan^{n-1} u - \int \tan^{n-2} u du$
43. $\int \cot^n u du = -\frac{1}{n-1}\cot^{n-1} u - \int \cot^{n-2} u du$
44. $\int \sec^n u du = \frac{1}{n-1}\sec^{n-2} u \tan u + \frac{n-2}{n-1} \int \sec^{n-2} u du$
45. $\int \csc^n u du = -\frac{1}{n-1}\csc^{n-2} u \cot u + \frac{n-2}{n-1} \int \csc^{n-2} u du$
46. $\int u \sin u du = \sin u - u \cos u + c$
47. $\int u \cos u du = \cos u + u \sin u + c$
67. $\int u^n \tan^{-1} u du = \frac{1}{n+1} \left(u^{n+1} \tan^{-1} u - \int \frac{u^{n+1} du}{1+u^2} \right),$
 $n \neq -1$
68. $\int \sinh u du = \cosh u + c$
69. $\int \cosh u du = \sinh u + c$
70. $\int \tanh u du = \ln |\cosh u| + c$
71. $\int \coth u du = \ln |\sinh u| + c$
72. $\int \operatorname{sech} u du = \tan^{-1}(\sinh u) + c$
73. $\int \operatorname{csch} u du = \ln \left| \tanh \frac{1}{2} u \right| + c$
74. $\int \operatorname{sech}^2 u du = \tanh u + c$
75. $\int \operatorname{csch}^2 u du = -\coth u + c$
76. $\int \operatorname{sech} u \tanh u du = -\operatorname{sech} u + c$
77. $\int \operatorname{csch} u \coth u du = -\operatorname{csch} u + c$
78. $\int \sinh^2 u du = \frac{1}{4}\sinh 2u - \frac{1}{2}u + c$
79. $\int \cosh^2 u du = \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$
80. $\int \tanh^2 u du = u - \tanh u + c$
81. $\int \coth^2 u du = u - \coth u + c$
82. $\int u \sinh u du = u \cosh u - \sinh u + c$
83. $\int u \cosh u du = u \sinh u - \cosh u + c$
84. $\int e^{au} \sinh nu du = \frac{e^{au}}{a^2 - n^2}(a \sinh nu - n \cosh nu) + c$
85. $\int e^{au} \cosh nu du = \frac{e^{au}}{a^2 - n^2}(a \cosh nu - n \sinh nu) + c$
86. $\int \frac{udu}{a+bu} = \frac{1}{b^2}(a+bu - a \ln |a+bu|) + c$
87. $\int \frac{u^2 du}{a+bu} = \frac{1}{b^3} \left(\frac{1}{2}(a+bu)^2 - 2a(a+bu) + a^2 \ln |a+bu| \right) + c$
88. $\int \frac{udu}{(a+bu)^2} = \frac{1}{b^2} \left(\frac{a}{a+bu} + \ln |a+bu| \right) + c$
89. $\int \frac{u^2 du}{(a+bu)^2} = \frac{1}{b^3} \left(a+bu - \frac{a^2}{a+bu} - 2a \ln |a+bu| \right) + c$

$$48. \int u^2 \operatorname{sen} u \, du = 2u \operatorname{sen} u + (2 - u^2) \cos u + c$$

$$90. \int \frac{u \, du}{(a + bu)^3} = \frac{1}{b^2} \left(\frac{a}{2(a + bu)^2} - \frac{1}{a + bu} \right) + c \quad 91. \int \frac{du}{u(a + bu)} = \frac{1}{a} \ln \left| \frac{u}{a + bu} \right| + c$$

$$92. \int \frac{du}{u^2(a + bu)} = -\frac{1}{au} + \frac{b}{a^2} \ln \left| \frac{a + bu}{u} \right| + c \quad 93. \int \frac{du}{u(a + bu)^2} = \frac{1}{a(a + bu)} + \frac{1}{a^2} \ln \left| \frac{u}{a + bu} \right| + c$$

$$94. \int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + c$$

$$95. \int \frac{du}{a^2 - u^2} = \frac{1}{2a} \ln \left| \frac{u + a}{u - a} \right| + c = \begin{cases} \frac{1}{a} \tanh^{-1} \frac{u}{a} + c, & \text{si } |u| < a \\ \frac{1}{a} \coth^{-1} \frac{u}{a} + c, & \text{si } |u| > a \end{cases}$$

$$96. \int \frac{du}{u^2 - a^2} = \frac{1}{2a} \ln \left| \frac{u - a}{u + a} \right| + c = \begin{cases} -\frac{1}{a} \tanh^{-1} \frac{u}{a} + c, & \text{si } |u| < a \\ -\frac{1}{a} \coth^{-1} \frac{u}{a} + c, & \text{si } |u| > a \end{cases}$$

$$97. \int u \sqrt{a + bu} \, du = \frac{2}{15b^3} (3bu - 2a)(a + bu)^{3/2} + c$$

$$98. \int u^2 \sqrt{a + bu} \, du = \frac{2}{105b^3} (15b^2u^2 - 12abu + 8a^2)(a + bu)^{3/2} + c$$

$$99. \int u^n \sqrt{a + bu} \, du = \frac{2u^n(a + bu)^{3/2}}{b(2n + 3)} - \frac{2an}{b(2n + 3)} \int u^{n-1} \sqrt{a + bu} \, du$$

$$100. \int \frac{u \, du}{\sqrt{a + bu}} = \frac{2}{3b^2} (bu - 2a) \sqrt{a + bu} + c$$

$$101. \int \frac{u^2 \, du}{\sqrt{a + bu}} = \frac{2}{15b^3} (3b^2u^2 - 4abu + 8a^2) \sqrt{a + bu} + c$$

$$102. \int \frac{u^n \, du}{\sqrt{a + bu}} = \frac{2u^n \sqrt{a + bu}}{b(2n + 1)} - \frac{2an}{b(2n + 1)} \int \frac{u^{n-1}}{\sqrt{a + bu}} \, du$$

$$103. \int \frac{du}{u \sqrt{a + bu}} = \begin{cases} \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a + bu} - \sqrt{a}}{\sqrt{a + bu} + \sqrt{a}} \right| + c, & \text{si } a > 0 \\ \frac{2}{\sqrt{-a}} \tan^{-1} \sqrt{\frac{a + bu}{-a}} + c, & \text{si } a < 0 \end{cases}$$

$$104. \int \frac{du}{u^n \sqrt{a + bu}} = -\frac{\sqrt{a + bu}}{a(n-1)u^{n-1}} - \frac{b(2n-3)}{2a(n-1)} \int \frac{du}{u^{n-1} \sqrt{a + bu}}$$

$$105. \int \frac{\sqrt{a + bu} \, du}{u} = 2\sqrt{a + bu} + a \int \frac{du}{u \sqrt{a + bu}}$$

$$106. \int \frac{\sqrt{a + bu} \, du}{u^n} = -\frac{(a + bu)^{3/2}}{a(n-1)u^{n-1}} - \frac{b(2n-5)}{2a(n-1)} \int \frac{\sqrt{a + bu} \, du}{u^{n-1}}$$

107. $\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln \left| u + \sqrt{u^2 \pm a^2} \right| + c$
108. $\int \sqrt{u^2 \pm a^2} \, du = \frac{u}{2} \sqrt{u^2 \pm a^2} \pm \frac{a^2}{2} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + c$
109. $\int u^2 \sqrt{u^2 \pm a^2} \, du = \frac{u}{8} (2u^2 \pm a^2) \sqrt{u^2 \pm a^2} - \frac{a^4}{8} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + c$
110. $\int \frac{\sqrt{u^2 + a^2} \, du}{u} = \sqrt{u^2 + a^2} - a \ln \left| \frac{a + \sqrt{u^2 + a^2}}{u} \right| + c$
111. $\int \frac{\sqrt{u^2 - a^2} \, du}{u} = \sqrt{u^2 - a^2} - a \sec^{-1} \frac{u}{a} + c$
112. $\int \frac{\sqrt{u^2 \pm a^2} \, du}{u^2} = -\frac{\sqrt{u^2 \pm a^2}}{u} + \ln \left| u + \sqrt{u^2 \pm a^2} \right| + c$
113. $\int \frac{u^2 \, du}{\sqrt{u^2 \pm a^2}} = \frac{u}{2} \sqrt{u^2 \pm a^2} \mp \frac{a^2}{2} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + c$
114. $\int \frac{du}{u \sqrt{u^2 + a^2}} = -\frac{1}{a} \ln \left| \frac{a + \sqrt{u^2 + a^2}}{u} \right| + c$
115. $\int \frac{du}{u \sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{u}{a} + c$
116. $\int \frac{du}{u^2 \sqrt{u^2 \pm a^2}} = \mp \frac{\sqrt{u^2 \pm a^2}}{a^2 u} + c$
117. $\int (u^2 \pm a^2)^{3/2} \, du = \frac{u}{8} (2u^2 \pm 5a^2) \sqrt{u^2 \pm a^2} + \frac{3a^4}{8} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + c$
118. $\int \frac{du}{(u^2 \pm a^2)^{3/2}} = \pm \frac{u}{a^2 \sqrt{u^2 \pm a^2}} + c$
119. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \frac{u}{a} + c$
120. $\int \sqrt{a^2 - u^2} \, du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \frac{u}{a} + c$
121. $\int u^2 \sqrt{a^2 - u^2} \, du = \frac{u}{8} (2u^2 - a^2) \sqrt{a^2 - u^2} + \frac{a^4}{8} \sin^{-1} \frac{u}{a} + c$
122. $\int \frac{\sqrt{a^2 - u^2}}{u} \, du = \sqrt{a^2 - u^2} - a \ln \left| \frac{a + \sqrt{a^2 - u^2}}{u} \right| + c = \sqrt{a^2 - u^2} - a \cosh^{-1} \frac{a}{u} + c$
123. $\int \frac{\sqrt{a^2 - u^2}}{u^2} \, du = -\frac{\sqrt{a^2 - u^2}}{u} - \sin^{-1} \frac{u}{a} + c$
124. $\int \frac{u^2}{\sqrt{a^2 - u^2}} \, du = -\frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \frac{u}{a} + c$
125. $\int \frac{du}{u \sqrt{a^2 - u^2}} = -\frac{1}{a} \ln \left| \frac{a + \sqrt{a^2 - u^2}}{u} \right| + c = -\frac{1}{a} \cosh^{-1} \frac{a}{u} + c$
126. $\int \frac{du}{u^2 \sqrt{a^2 - u^2}} = -\frac{\sqrt{a^2 - u^2}}{a^2 u} + c$

$$127. \int (a^2 - u^2)^{3/2} du = -\frac{u}{8}(2u^2 - 5a^2)\sqrt{a^2 - u^2} + \frac{3a^4}{8} \sin^{-1} \frac{u}{a} + c$$

$$128. \int \frac{du}{(a^2 - u^2)^{3/2}} = \frac{u}{a^2 \sqrt{a^2 - u^2}} + c$$

$$129. \int \sqrt{2au - u^2} du = \frac{u-a}{2} \sqrt{2au - u^2} + \frac{a^2}{2} \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$130. \int u \sqrt{2au - u^2} du = \frac{2u^2 - au - 3a^2}{6} \sqrt{2au - u^2} + \frac{a^3}{2} \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$131. \int \frac{\sqrt{2au - u^2}}{u} du = \sqrt{2au - u^2} + a \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$132. \int \frac{\sqrt{2au - u^2}}{u^2} du = -\frac{2\sqrt{2au - u^2}}{u} - \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$133. \int \frac{du}{\sqrt{2au - u^2}} = \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$134. \int \frac{udu}{\sqrt{2au - u^2}} = -\sqrt{2au - u^2} + a \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$135. \int \frac{u^2 du}{\sqrt{2au - u^2}} = -\frac{u+3a}{2} \sqrt{2au - u^2} + \frac{3a^2}{2} a \cos^{-1} \left(1 - \frac{u}{a} \right) + c$$

$$136. \int \frac{du}{u \sqrt{2au - u^2}} = -\frac{\sqrt{2au - u^2}}{au} + c$$

$$137. \int \frac{du}{(2au - u^2)^{3/2}} = \frac{u-a}{a^2 \sqrt{2au - u^2}} + c$$

$$138. \int \frac{udu}{(2au - u^2)^{3/2}} = \frac{u}{a \sqrt{2au - u^2}} + c$$